

Assignment Report

Title: **Compiler Principles**

The Eighth Theory Homework

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[ABSTRACT]

After learning the lecture 8 of compiler principles[1], we got the homework at the end of the lecture.

There are some requirements for the homework:

- The file should be named as **“学号 _ 姓名 _ 第 i 次理论作业”**, where i is the number of the homework.
- The file should be in docx or **pdf** format.
- All the homework should be sent to the TA's email: **406648896@qq.com**
- The deadline for the homework is **24:00 on Sunday of the week** when the homework is assigned.

Keywords: Compiler Principles

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1 Exercise 8

1.1 Exercise 8.1

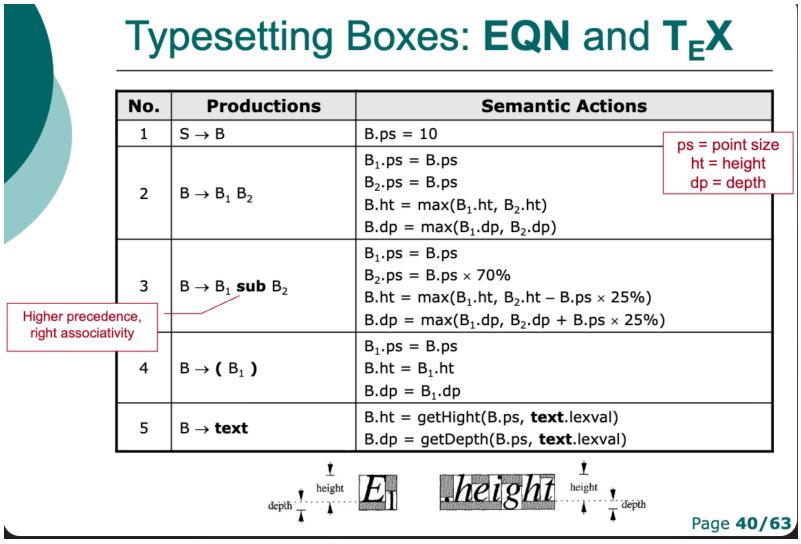
1.1.1 Problem

Given the translation scheme for the EQN language (see pp. 40 in this lecture), calculate the height and depth of the input:

```
text sub text sub text.
```

Suppose that for each `text`,

- getHeight(ps, `text.lexval`) $= 8 \times ps$,
- getDepth(ps, `text.lexval`) $= 0$.



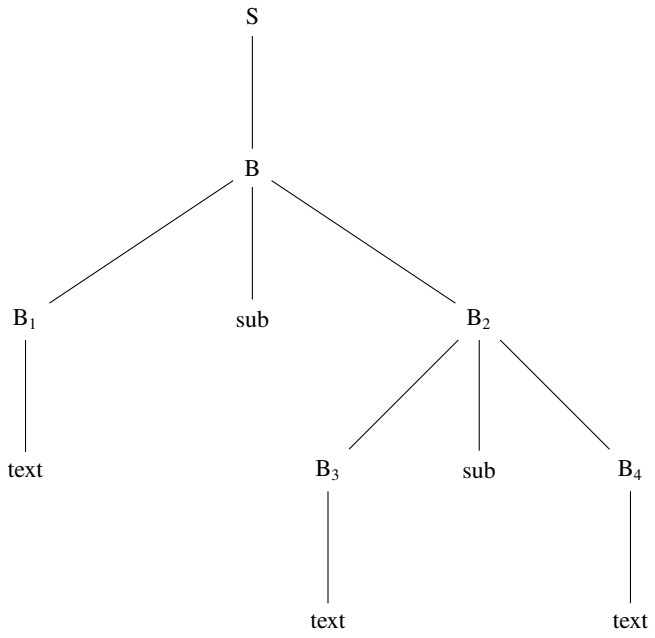
1.1.2 Solution

We compute attributes bottom-up using the semantic rules on Page 40/63.

The input `text sub text sub text` is parsed as right-associative:

```
text sub (text sub text)
```

Corresponding parse tree (nonterminals labeled as in the grammar):



Now apply semantic actions:

1) Rule 1: $S \to B$ gives $B.ps = 10$.
2) Rule 3 ($B \to B_1$ `sub` B_2):
   - $B_1.ps = B.ps = 10$,
   - $B_2.ps = B.ps \times 70\% = 10 \times 0.7 = 7$.
3) Rule 5 ($B_1 \to$ `text`):
   - $B_1.ht = 8 \times B_1.ps = 8 \times 10 = 80$,
   - $B_1.dp = 0$.
4) Rule 3 applied to $B_2 \to B_3$ `sub` B_4:
   - $B_3.ps = B_2.ps = 7$,
   - $B_4.ps = B_2.ps \times 70\% = 7 \times 0.7 = 4.9$.
5) Rule 5 for leaves:
   - $B_3.ht = 8 \times 7 = 56, \quad B_3.dp = 0$,
   - $B_4.ht = 8 \times 4.9 = 39.2, \quad B_4.dp = 0$.

6) Rule 3 for B_2 (synthesized attributes):

$$\begin{aligned} B_2.ht &= \max\big(B_3.ht,\ B_4.ht - B_2.ps \times 25\%\big) \\ &= \max\big(56,\ 39.2 - 7 \times 0.25\big) = \max(56,\ 39.2 - 1.75) = \max(56,\ 37.45) = 56, \\ B_2.dp &= \max\big(B_3.dp,\ B_4.dp + B_2.ps \times 25\%\big) \\ &= \max\big(0,\ 0 + 1.75\big) = 1.75. \end{aligned}$$

7) Finally, Rule 3 for the top-level B:

$$\begin{aligned} B.ht &= \max\big(B_1.ht,\ B_2.ht - B.ps \times 25\%\big) \\ &= \max\big(80,\ 56 - 10 \times 0.25\big) = \max(80,\ 53.5) = \boxed{80}, \\ B.dp &= \max\big(B_1.dp,\ B_2.dp + B.ps \times 25\%\big) \\ &= \max\big(0,\ 1.75 + 10 \times 0.25\big) = \max(0,\ 4.25) = \boxed{4.25}. \end{aligned}$$

Thus, the height of the expression is 80 and the depth is 4.25.

References

[1] JUN LI W. Lecture08. principles of compiler construction[Z]. 2026.